

Kai Hori

Davis, CA | horipapa0613@gmail.com | (413) 409-4358 | www.linkedin.com/in/kai-hori | <https://kaihor1613.github.io/>

EDUCATION

University of California, Davis

Bachelor of Science, Mechanical Engineering

GPA: 3.614/4.0

Bachelor of Science, Aerospace Science and Engineering

Relevant Coursework: Mechanics of Materials, Fluid Mechanics, Thermodynamics, Heat Transfer, Automatic Control of Engineering Systems, Aircraft Propulsion, Applied Aerodynamics, Compressible Aerodynamics, Rocket Propulsion, and Space Vehicle Design

TECHNICAL SKILLS

CAD & Engineering Software: Onshape, SOLIDWORKS, Fusion 360, ANSYS Fluent, NASTRAN/PATRAN

Programming & Analysis: MATLAB, Simulink, Python, C#, Fortran, Unity, HLSL

Manufacturing & Prototyping: 3D Printing, Manual Milling, Lathe, CNC Milling, Drilling, Welding

Documentation: Microsoft Excel, Word, PowerPoint, LaTeX, and Google Workspace

Languages: English (Fluent), Japanese (Fluent)

ENGINEERING AND RESEARCH EXPERIENCE

Research Assistant - *Human/Robotics/Vehicle Integration & Performance Laboratory*

Feb 2026 - Present

- Design, fabricate, assemble, and test ground-station hardware for the MOON initiative, including QFH antennas, parabolic-dish feed systems, mounting components, and satellite-tracking equipment.
- Conduct iterative field testing and hardware troubleshooting to evaluate mechanical alignment, mounting configuration, antenna positioning, and environmental effects on system performance.

Research Intern - *NeuralX*

July 2025 - September 2025

- Developed Smoothed Particle Hydrodynamics simulations in Unity to model water-like fluid behavior.
- Improved computational performance through GPU parallelization, spatial-neighbor searching, and compute-shader optimization while evaluating numerical stability and simulated fluid behavior.

Structure Lead - *Truss-Based Wing Structural & Aeroelastic Analysis Class Project*

April 2026 - June 2026

- Developed Patran/NASTRAN finite-element models of a truss-braced wing and performed static and modal analyses to evaluate structural behavior, natural frequencies, and mode shapes.
- Used MATLAB and flutter-analysis results to assess aeroelastic stability, including flutter and divergence behavior across the operating envelope.

Communications Lead - *Spacecraft Inspection Mission Design Project, Senior Capstone Class*

April 2026 - June 2026

- Led multidisciplinary engineering analyses and system-level trade studies for a heliocentric spacecraft design, coordinating technical interfaces across communications, power, payload, and spacecraft operations.
- Developed MATLAB-based performance models, system requirements, verification plans, and failure-mode analyses to support system integration and design decisions.

Payload Lead - *Spacecraft Inspection Mission Redesign Project Senior Capstone Class*

January 2026 - March 2026

- Redesigned NASA's CPOD CubeSat concept for autonomous, non-docking spacecraft inspection, evaluating LiDAR integration, sensor performance, mass, power, and onboard-processing constraints.

Research Assistant - *UC Davis Computational Flow Physics & Aeroacoustics Laboratory*

Dec 2024 - Apr 2025

- Used Fortran and Microsoft Visual Studio to configure computational cases, troubleshoot software implementation, execute simulations, and interpret numerical results.

OTHER WORK EXPERIENCE

Tutor - *UC Davis Academic Assistance and Tutoring Centers*

April 2025 - April 2026

- Provide academic assistance to undergraduate students in Electromagnetism, Calculus, Dynamics, and Linear Algebra through both drop-in and individual tutoring sessions.

Co-Instructor - *UC Davis International Center*

June 2024 - June 2026

- Facilitated and co-taught business fundamentals to agriculture students, translated core concepts (finance, operations, marketing) into industry-specific applications, demonstrating the ability to tailor solutions to a target audience.
- Partnered with a lead instructor as a culture intermediary and liaison to build a business course for the Japanese Agricultural Training Program in partnership with the Japanese Agricultural Exchange Council.